

Question1: Answer the following:

(6 + 5 M)

1. A computer science student is working on an algorithm that computes the square root of numbers. However, the algorithm produces approximations instead of exact values. The table below shows a few computed results:

Input x	True value $\sqrt{x}$	Computed value $\sqrt{x}$
4	2.0000	2.1000
10	3.1623	3.1600

- a) Give the definitions of the absolute error and the relative error.
b) For each input, calculate the absolute error between the true value and the computed value.
c) For each input, calculate the relative error as a percentage.

a) Absolute error = | Exact value – Approximate value |
Relative error = | Exact value – Approximate value | / | Exact value |.

b) For x = 4: Absolute error = | 2.0000 – 2.1000 | = 0.1

For x = 10: Absolute error = | 3.1623 – 3.1600 | = 0.0023

c) For x = 4: Relative error = | 0.1 | / | 2.000 | = 0.05 = 5%

For x = 10: Relative error = | 0.0023 | / | 3.1623 | = 7.27×10^{-4} = 0.073%

2. If $f(x) = x^3 - 2x - 1$ and $x_0 = 1.5$, what are x_1 and x_2 in the Newton iteration.

From the basic formula, $x_{n+1} = x_n - f(x_n)/f'(x_n)$.

Now, $f'(x) = 3x^2 - 2$, **1**

Hence, we have: $x_1 = x_0 - f(x_0)/f'(x_0) = 1.5 - (-0.625/4.75) = 1.6316$. **2**

Similarly, $x_2 = x_1 - f(x_1)/f'(x_1) = 1.6316 - 0.0803/5.9864 = 1.6182$. **2**

Question 2: Floating-point representation**(6 m)**

Determine the single-precision machine representation of the decimal number 1259.125.

Solution:

1. Converting the integer part to binary, we have:

$$(1259)_{10} = (2353)_8 = (100\ 1110\ 1011)_2. \quad 1$$

2. Converting the fractional part to binary, we have:

$$(.125)_{10} = (.1)_8 = (.001)_2. \quad 1$$

- 3.
- $(1259.125)_{10} = (100\ 1110\ 1011.001)_2 = (1.0011101011001)_2 \times 2^{10}$
- is the corresponding one-plus form in base 2, and
- $(.0011101011001)_2$
- is the stored mantissa. 1

4. Next the exponent is
- $(10)_{10}$
- , then
- $c - 127 = 10$
- , we immediately see that
- $(137)_{10} = (211)_8 = (10001001)_2$
- is the stored exponent. 2

Thus, the single-precision machine representation of 1259.125 is:

$$[0100010010011101011001000000000000]_2$$

$$[0100\ 0100\ 1001\ 1101\ 0110\ 0100\ 0000000000]_2 = [44\ 9D6400]_{16} \quad 1$$

Question 3: Given the following function:**(2+6 M)**

$$f(x) = e^x - 4x^2$$

1. Prove that the function f has exactly one real root α in the interval $[0; 1]$.
2. Find a root of the function $f(x)$ on the interval $[0; 1]$ by using the bisection method. Terminate the iterations if they exceed $n_{max} = 2$ or if the error bound is less than 0.5×10^{-6} .

Solution:

$$1. \quad u = f(0) = 1$$

$$v = f(1) = e^1 - 4 \cdot 1^2 = 2.14 - 4 = -1.2817$$

Since, $u \cdot v < 0$ then the function f has exactly one real root α in the interval $[0; 1]$.

2.

$$1) \quad n = 0 \text{ and } c = (0 + 1)/2 = 1/2, \quad w = f(c) = f(0.5) = 0.6487$$

u.w > 0 then: a = c, b = b and the interval $[0.5, 1]$. Error = 0.5. 2

$$2) \quad n = 1 \text{ and } c = (0.5 + 1)/2 = 0.75 \quad w = f(0.75) = -0.1330$$

u.w < 0 then: a = a, b = c and the interval $[0.5, 0.75]$. Error = 0.25. 2

$$3) \quad n = 2 \text{ and } c = (0.5 + 0.75)/2 = 0.625 \quad w = f(0.625) = 0.3057$$

u.w > 0 then : a = c , b = b and the interval $[0.625, 0.75]$. Error = 0.125. 2

n	cn	$f(cn)$	Error
0	0.5	0.6487	0.5
1	0.75	-0.1330	0.25
2	0.625	0.3057	0.125